Zenith Saru

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PROFESSIONAL SUMMARY

Civil Engineering graduate (Thapathali Campus, IOE, TU) with practical GIS and remote sensing experience alongside structural and water resources engineering. Experienced in satellite image classification, land cover change detection, terrain and watershed analysis, InSAR deformation mapping, and map production using QGIS, Google Earth Engine, and Python. Teaches a paid hands-on QGIS workshop for civil engineering students. Also skilled in structural analysis and design, BOQ and cost estimation, and hydraulic modeling with HEC-RAS. Proven leadership as CESS President, driving teams from concept to execution.

EDUCATION

Bachelor of Engineering in Civil Engineering *Year of Completion: 2026 AD*

Thapathali Campus, Institute of Engineering (IOE), Tribhuvan University

TECHNICAL SKILLS

Software & Tools:
AutoCAD, ETABS, QGIS, Python, Microsoft Office Suite (Excel, Word, PowerPoint, Access)

GIS & Remote Sensing:
QGIS, Google Earth Engine, Landsat and Sentinel imagery, supervised land cover classification, change detection, DEM processing, watershed delineation and stream network extraction, InSAR time series (LiCSBAS, GACOS), spatial statistics (Sen's slope, Getis-Ord Gi*), map layout and cartography

Programming & Analysis:
Python, Machine Learning (Random Forest, XGBoost), MATLAB

Structural Design:
RCC and Steel design of several residential and commercial buildings using ETABS, SAP2000, IdeaStatica, SAFE and AutoCAD

Estimation & Costing:
Quantity take-off, BOQ preparation, and cost report estimation for RCC residential structures

Languages:
English (Fluent), Nepali (Native), Hindi (Conversational)

EXPERIENCE

Civil Engineering Intern – Dwarika Construction and Engineering Pvt. Ltd. – Baneshwor, Kathmandu:
(From December 2025 to July 2026)

- Prepared BOQs and estimates for 5 projects totaling NPR 10 crore+, ranging from steel-frame warehouses and RCC cold stores to colleges and hostels
- Reviewed structural analyses and drawings, and prepared the administrative and client documentation needed for project approvals

Structural Engineer – Bela Industries Nepal Pvt. Ltd. – Chhauni, Kathmandu
(From July 2026 – Current)

- Delivered structural analysis for 14 buildings, including a 4-storey steel-frame and EPS-panel school, by modeling in ETABS from architectural drawings and client plans
- Prepared BOQs and estimates worth NPR 25 crore+ for civil works
- Conducted periodic site visits to check construction against drawings, specifications, and agreed timelines

- Resolved client structural and construction queries, turning engineering decisions into clear, actionable guidance

GIS Instructor – QGIS Workshop – Kathmandu

- Trained 50+ civil engineering students in practical GIS through a paid 10-day workshop, taking them from DEM processing to full watershed delineation on the Manohara River basin
- Built 10 theory decks with matching hands-on labs covering flow direction, flow accumulation, Strahler stream order, pour points, and satellite imagery download
- Enabled 30+ students to produce properly formatted GIS maps and reports for their academic submissions

ACADEMIC PROJECTS

Forest and Snow Cover Dynamics in Nepal: Predictive Modeling and Hotspot Detection | Final Year Project

- Mapped 14 years of forest and snow change across Nepal at 97.83% accuracy (Kappa = 0.96) by classifying Landsat 5/8/9 imagery with Random Forest on Google Earth Engine
- Found a 44.9% snow cover decline from 2011 to 2025 and flagged the Mahakali and Karnali basins as extreme-risk zones using a composite hotspot index (Sen's slope, Getis-Ord Gi*)
- Projected forest and snow cover to 2100 under SSP2-4.5 and SSP5-8.5 by building a hybrid XGBoost–Random Forest model driven by downscaled CMIP6 data

Kathmandu Valley Land Subsidence Mapping using InSAR | Ongoing Research

- Processing Sentinel-1 InSAR time series with LiCSBAS and GACOS atmospheric correction to map ground deformation across Kathmandu Valley
- Combining ascending and descending tracks to separate vertical and east-west displacement

Eco-Industrial Park Model | Civil Model Competition

- Conceptualized and led team in developing an industrial symbiosis model integrating steel factory, water treatment, agriculture, cement manufacturing, and settlement
- Demonstrated circular economy principles by designing waste-to-resource conversion systems where one industry's waste serves as raw material for another
- Conducted initial research, formed project team, and designated roles for successful competition participation

LEADERSHIP EXPERIENCE

President, Civil Engineering Students' Society (CESS) | 2082 BS

- Led a 25-member executive team to deliver 30+ events in one year, including the campus's flagship annual event with renowned industry professionals and TV journalists as moderators
- Spearheaded panel discussions on critical topics including Disaster Risk Management, Water Supply & Hydropower Infrastructure, and Professional Ethics in Construction

Exhibition Coordinator, Utsarga 078, Yathartha 1.0 & 2.0 | National Level Engineering Exhibition

- Managed exhibition logistics and coordination for national-level engineering showcase events

Editorial Team, Civil Transcend Magazine

- Editor, Issue II - Reviewed and curated technical content for student engineering publication
- Coordinator, Issue III - Led editorial team and managed publication workflow

PUBLICATIONS

"Forest and Snow Cover Dynamics in Nepal: Predictive Modelling and Hotspot Detection Using Machine Learning" | Journal of Mountain Science (Under Revision)

- Converted our final year project into a research-grade paper and submitted it to a Q2 Journal with IF of 2.5+
- Reviewed and cited more than 80+ literature during the process of writing the manuscript

"Sustainability of Hydropower Projects in Nepal across three basins: Naumure, Trishuli, and Arun under different Climatic Scenarios" | Journal of Innovations in Engineering Education (Under Review)

- Compiled the final year project reports of all the water engineering related projects and conducted a symmetric comparison of the Q40 power projection of all the hydropower projects within the basins.
- Familiarized with the hydropower and climatic conditions correlations in terms of power generation and sustainability.